

Challenge (Habanero)

- Q1.** A hiker needs to cover $3x + 2y \leq 12$ miles in a forest, where x is the distance traveled east and y is the distance traveled north. Which of the following is a feasible solution?
- (A) $x = 2, y = 1$
 (B) $x = 3, y = 2$
 (C) $x = 1, y = 3$
 (D) $x = 4, y = 0$
 (E) $x = 0, y = 5$
- Q2.** Given the system of inequalities $x + y \geq 2$ and $x - y \leq 1$, what is the overlapping solution region?
- (A) $x \geq 1, y \geq 1$
 (B) $x \leq 1, y \leq 1$
 (C) $x \geq 1, y \leq 1$
 (D) $x \leq 1, y \geq 1$
 (E) $x = 1, y = 1$
- Q3.** The population P of a certain animal species can be modeled by the inequalities $P \geq 100$ and $P \leq 500$. What is the range of possible populations?
- (A) $0 \leq P \leq 100$
 (B) $100 \leq P \leq 500$
 (C) $500 \leq P \leq 1000$
 (D) $0 \leq P \leq 500$
 (E) $100 \leq P \leq 1000$
- Q4.** A group of friends want to hike a trail that is x miles long, where $x + 2y \leq 10$ and $x - 2y \geq -4$. Which of the following is a feasible solution?
- (A) $x = 2, y = 2$
 (B) $x = 4, y = 1$
 (C) $x = 1, y = 3$
 (D) $x = 6, y = 0$
 (E) $x = 0, y = 4$
- Q5.** The amount of fuel F consumed by a car is given by the inequalities $F \geq 10$ and $F \leq 30$. What is the range of possible fuel consumption?
- (A) $0 \leq F \leq 10$
 (B) $10 \leq F \leq 30$
 (C) $30 \leq F \leq 50$
 (D) $0 \leq F \leq 30$
 (E) $10 \leq F \leq 50$
- Q6.** A hiker needs to cover a distance of x miles, where $2x + y \leq 12$ and $x - 2y \geq -4$. Which of the following is a feasible solution?
- (A) $x = 2, y = 2$
 (B) $x = 4, y = 1$
 (C) $x = 1, y = 3$
 (D) $x = 6, y = 0$
 (E) $x = 0, y = 4$
- Q7.** The population P of a certain animal species can be modeled by the inequalities $P \geq 50$ and $P \leq 200$. What is the range of possible populations?
- (A) $0 \leq P \leq 50$
 (B) $50 \leq P \leq 200$
 (C) $200 \leq P \leq 500$
 (D) $0 \leq P \leq 200$
 (E) $50 \leq P \leq 500$
- Q8.** Given the system of inequalities $x + 2y \geq 4$ and $x - 2y \leq 2$, what is the overlapping solution region?
- (A) $x \geq 2, y \geq 1$
 (B) $x \leq 2, y \leq 1$
 (C) $x \geq 2, y \leq 1$
 (D) $x \leq 2, y \geq 1$
 (E) $x = 2, y = 1$

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities $x + y \leq 4$ and $x - y \geq -2$. Shade the feasible region and identify the corner points.
- Q10.** A car's fuel consumption F is given by the inequalities $F \geq 15$ and $F \leq 35$. Graph the inequalities and shade the feasible region. What is the range of possible fuel consumption?

Chef's Tips

- Tip 1: Ensure students understand the concept of feasible regions and corner points.
- Tip 2: Emphasize the importance of accurate graphing and shading.
- Tip 3: Encourage students to check their work and verify their answers.